

Lecture 8: POS Tagging & Sentiment Analysis

Week 2: From Structure to Meaning

PSYC 51.07: Models of Language and Communication

Learning Objectives

By the end of this lecture, you will:

1. Understand part-of-speech (POS) tagging and its applications
2. Explore how neural networks learn grammatical structure
3. Apply sentiment analysis to real-world text
4. Fine-tune pre-trained models for domain-specific tasks
5. Critically evaluate whether models "understand" language

Central questions:

- Can statistical patterns capture grammatical knowledge? 🤔
- What does it mean for a model to "understand" emotion? 😊😞

Part-of-Speech (POS) Tagging

****What is POS tagging?****

– Assigning grammatical category to each word

- Categories: noun, verb, adjective, adverb, pronoun, preposition, etc.
- A fundamental NLP task

Example:

1The -> cat -> sat -> on -> the -> mat -> DET -> NOUN

****Why it matters:****

– Disambiguation: "book" as noun vs. verb

- Syntax parsing and understanding

POS Tagsets: Universal vs. Fine-Grained

```
<div class="columns">
```

Universal POS Tags (17 tags):

- ADJ: adjective
 - ADP: adposition
 - ADV: adverb
 - AUX: auxiliary
 - CONJ: conjunction
 - DET: determiner
 - NOUN: noun
 - NUM: numeral

Context Matters: Ambiguous Words

****Many words have multiple possible POS tags:****

Sentence	Word	POS	Explanation
"I read a book "	book	NOUN	Object being read
"Please book a table"	book	VERB	Action of reserving
"She runs fast "	fast	ADV	Modifies "runs"
"I will fast today"	fast	VERB	Action of not eating
"Please close the door"	close	VERB	Action
"Stay close to me"	close	ADV	Modifies position

****Key insight:**** Context determines POS! Models must look at surrounding words.

Ambiguity in Action: Code Example

How spaCy resolves ambiguity based on context:

```
1import spacy
2nlp = spacy.load("en_core_web_sm")
3
4sentences = [
5    "I need to book a flight",      # book = VERB
6    "I'm reading a great book",    # book = NOUN
7    "The record was broken",        # record = NOUN
8    "Please record the meeting",     # record = VERB
9]
10
11for sent in sentences:
12    doc = nlp(sent)
13    for token in doc:
14        if token.text.lower() in ["book", "record"]:
15            print(f'"{sent}"')
16            print(f'  "{token.text}" -> {token.pos} ({spacy.explain(token.pos)})')
```

Ambiguity in Action: Code Example

```
21#    "book" -> VERB (verb)
22#
23# "I'm reading a great book"
24#    "book" -> NOUN (noun)
```

...continued

The model uses surrounding words ("to book" vs "a book") to disambiguate!

POS Tagging with spaCy

```
```python
```

```
import spacy
```

```
nlp = spacy.load("en_core_web_sm")
```

```
sentence = "She will book the meeting room tomorrow"
```

```
doc = nlp(sentence)
```

```
print(f"{'Word':<12} {'POS':<8} {'Detailed Tag':<12} {'Explanation'}")
```

```
print("-" * 55)
```

```
for token in doc:
```

```
 explanation = spacy.explain(token.pos_)
```

```
 print(f"{token.text:<12} {token.pos_:<8} {token.tag_:<12} {explanation}")
```



## tomorrow NOUN NN noun, singular or mass

```
21
22
23 **Advantages of neural methods:**
24 - Learn patterns from data (no hand-crafted rules)
25- Capture long-range dependencies
26- Handle ambiguity through context
27- State-of-the-art accuracy (>97% on English!)
28
29
30---
31
32# Neural Networks and Grammar 🤖
33
34
35 **Can neural networks learn syntactic structure?**
36
37
38
```

# tomorrow NOUN NN noun, singular or mass

```
41
42
43<!-- DIAGRAM: TikZ conversion needed -->
44<!-- Original TikZ code preserved for reference:
45
46
47
48
49
50 \node[left, font=, align=right] at (-3.7,1.5) {Mu...
51-->
52
```

...continued

[Diagram placeholder - manual conversion required]

# tomorrow NOUN NN noun, singular or mass

```
21
22# BLiMP: Testing Linguistic Knowledge 🇫🇷
23
24
25 **BLiMP Dataset** (Warstadt et al., 2020):
26 - Benchmark of Linguistic Minimal Pairs
27- 67,000 sentence pairs across 67 paradigms
28- Tests syntax, semantics, morphology
29
30
31
32 **Example minimal pairs:**
33
34
35 | "Who did you see?" | "Who did you saw?" |
36| --- | --- |
37| "I think that she left" | "I think that she leave" |
38
```

# tomorrow NOUN NN noun, singular or mass

41  
42  
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53---  
54  
55#  
56  
57

**\*\*Task:\*\*** Model should assign higher probability to acceptable sentence

**\*\*Results:\*\*** Transformers (BERT, GPT-2) perform well (70–85%), but not per

 **\*\*Reference:\*\*** Warstadt et al. (2020). BLiMP: The Benchmark of Linguist

Token Classification with HuggingFace 🤗

...

21

22

**\*\*Your thoughts?\*\*** Is pattern matching sufficient for grammatical competen

23

24---

25

26# Sentiment Analysis: From Structure to Emotion 😊😞

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28

29 **\*\*What is sentiment analysis?\*\***

30 - Determining emotional tone of text

31- Classification task: positive, negative, neutral (or fine-grained)

32- Moves beyond structure to *\*meaning\**

33

34

35

36 **\*\*Applications:\*\***

37 - 📱 Social media monitoring

38- 🛍️ Customer feedback analysis

...

```
41- 📦 Political opinion tracking
42
43
44
45 **Granularity levels:**
46 - Binary: positive vs. negative
47- Ternary: positive, negative, neutral
48- Fine-grained: 1-5 stars, or continuous score
49
50
51---
52
53# Sentiment Analysis Challenges 🚧
54
55
56 **1. Sarcasm and irony:**
57 - "Oh great, another meeting" (negative, despite "great")
58- "This is the best movie I've ever fallen asleep to" (negative!)
```

...

```
61
62 **2. Context-dependent sentiment:**
63 - "This movie is sick!" (positive in slang, negative literally)
64- "The book was long" (neutral? negative?)
65
66
67
68 **3. Mixed sentiment:**
69 - "Great food but terrible service" (both positive and negative)
70- Aspect-based sentiment: food=positive, service=negative
71
72
73
74 **4. Negation:**
75 - "not good" vs. "good"
76- "I don't dislike it" (double negative = positive?)
77
78
```

...

```
81 - "Explosive growth" (positive in business, negative in safety)
82- Different domains have different sentiment patterns
83
84
85---
86
87# Sentiment Challenges: Code Examples 🔍
88
89**Testing how models handle tricky cases:**
90
```

...continued

```
from transformers import pipeline sentiment = pipeline("sentiment-analysis")
```

```
tricky_cases = [
```

```
Sarcasm
```

```
("Oh great, another Monday meeting", "NEGATIVE/Unexpected")
```



...

```
21
22 | great | +3 |
23| --- | --- |
24| good | +3 |
25| hate | -3 |
26| terrible | -3 |
27| awful | -3 |
28
29
30
31
32
33 **Limitations:** Ignores context, word order, sarcasm...
34
35---
36
37# Lexicon-Based Sentiment Analysis 
38
```

# Sentiment: **NEGATIVE**, Confidence: 0.998

```
21| --- | --- | --- |
22| Twitter | TwitterBERT | Social monitoring |
23| Product Reviews | RoBERTa + Amazon | E-commerce |
24| Movies | BERT + IMDb | Film reviews |
25
26
27
28
29
30 **Why it works:**
31 - Domain-specific vocabulary ("bullish" in finance = positive)
32- Different sentiment expressions
33- Adapted to domain conventions
34
35
36---
37
38# Comparing General vs. Domain-Specific 
```

# Financial model often more accurate for finance text!

```
215. **Evaluate on test set**
22 - Accuracy, precision, recall, F1-score
23
24 📖 [HuggingFace Chapter 3.2: Processing Data for Fine-tuning](https://hugg
25
26---
27
28# Fine-Tuning Example (Simplified) 🔧
29
30
```

...continued

```
from transformers import AutoModelForSequenceClassification, Trainer
```

## 1. Load pre-trained model

## 4. Create Trainer and train

```
21 \]
22
23
24
25 **Why not just accuracy?**
26 - Imbalanced datasets (e.g., 90% positive reviews)
27- Different costs for false positives vs. false negatives
28- F1 gives balanced view of model performance
29
30
31---
32
33# Confusion Matrix Example 🤖👤
34
35
36
37
38<!-- DIAGRAM: TikZ conversion needed -->
```

## 4. Create Trainer and train

```
41 % Matrix
42
43
44
45
46 % Labels
47
48
49
50
51
52 % Values
53 ...
54-->
55
```

...continued

## 4. Create Trainer and train

```
21
22
23 **Aspect-based approach:**
24 - Food: Positive
25- Service: Negative
26- Atmosphere: Neutral
27
28
29
30 **Applications:**
31 - Restaurant reviews: food, service, ambiance, price
32- Product reviews: quality, price, shipping, customer service
33- Hotel reviews: room, location, staff, cleanliness
34
35
36
37 **More nuanced than overall sentiment!** Provides actionable insights.22
38
```

## 4. Create Trainer and train

```
41# Aspect-Based Sentiment: Worked Example 
42
43**Extracting aspect-level sentiment from a restaurant review:**
44
```

...continued

```
review = """ The pasta was absolutely incredible - best I've had in years! However, we
waited 45 minutes for our food which was frustrating. The ambiance was nice but a bit loud
for conversation. Prices were reasonable for the quality. """
```

### Manual aspect extraction (simplified approach)

```
aspects = {
 "food": ["pasta", "incredible", "best"], # POSITIVE
```

## Overall: 3/4 positive aspects -> Generally positive with service issues

```
21- Domain-specific model (if available)
223. **Analyze results:**
23 - Where do models disagree?
24- Which handles sarcasm better?
25- Which is most accurate for your domain?
264. **Bonus:** Fine-tune a model on your specific dataset!
27
28
29---
30
31# Discussion: Understanding Emotion 🤔
32
33
34 **Philosophical questions:**
35
36 1. **Can models "feel" sentiment?**
```



## Overall: 3/4 positive aspects -> Generally positive with service issues

```
413. **Cultural and linguistic variation:**
42 - Sentiment expressions vary across cultures
43- Can models capture these nuances?
444. **Ethical considerations:**
45 - Automated sentiment analysis in hiring, lending...
46- Risks of bias and discrimination
47- Should we trust model judgments?
48
49
50---
51
52# Week 2 Integration: Putting It All Together 🧩
53
54
55 **How the pieces connect:**
56
```